SURT AI

ML Engineering Internship | Progress Report Four

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Goals

Weeks 13–16: Rust–Python Interfacing
Implement core inference in Rust while calling open-source Python models where necessary (via pyo3, cxx, or gRPC). Experiment with multi-stage inference: e.g., motion tracking + depth estimation + face embedding. Deliverable: working demo calling a Python model from Rust.

Data

I evaluated the liveness detection pipeline using the Anti-Spoofing dataset from Kaggle: https://www.kaggle.com/datasets/tapakah68/anti-spoofing . It is a labeled face presentation attack detection (PAD) dataset containing both genuine (live) and spoofed facial media. The dataset includes both static images and video clips. The folder structure consists of:

- live_selfie (static images of real users)
- live_video (videos of real users)
- printouts (static printed photo attacks)
- cut-out printouts (printed photos presented within video)
- replay (screen replay video attacks)

Live samples are labeled as 1 and spoof samples as 0, with labels inferred directly from folder names. The dataset includes JPEG/PNG images and MP4/AVI/MOV videos, enabling evaluation of both spatial and temporal liveness cues. This makes it well-suited for testing a multi-stage inference pipeline.

This dataset was selected because it contains multiple common spoof modalities, supports both image and video evaluation, and integrates cleanly into a Rust based batch inference pipeline.

CASIA-FASD was not used for this milestone because it is not directly publicly downloadable and requires formal access approval and licensing. Since the objective of this phase was to validate system architecture, I thought that a more readily accessible dataset was better for faster iteration and integration testing.

Model

The liveness detection system is a modular, multi-stage inference pipeline made up of four independent signal extraction components followed by a weighted fusion and decision layer.

Spatial Stage (CNN Backbone)
The spatial stage uses a pretrained ResNet18 model (ImageNet weights) applied to a cropped facial region. For each frame, the maximum softmax probability is extracted as a proxy for structural confidence and image clarity. It is not fine-tuned for liveness, but this stage provides a learned feature-based signal reflecting facial structure consistency. For video inputs, frame-level scores are averaged.

I initially trialed FaceNet (InceptionResnetV1 pretrained on VGGFace2) as the spatial backbone but, it was unsuitable for liveness detection. FaceNet is trained for face identity recognition and produces consistently high confidence embeddings for both live and spoofed faces, as long as the face is recognizable. As a result, it lacked discriminative power for presentation attack detection. ResNet18, while not liveness-specific, gave more incidental separation across spoof types in this dataset and was then used as the spatial backbone for this project. A liveness-fine-tuned model will replace this stub in future iterations.

Texture Stage (Laplacian Variance)
The texture stage calculates the variance of the Laplacian on the cropped grayscale face region. This captures high frequency micro texture information which tends to be richer in live skin and reduced in flat print or screen based spoof attacks. The variance is normalized to a bounded [0,1] range using an empirical scaling constant.

Motion Stage (Inter-Frame Differencing)
The motion stage measures weighted pixel differences between consecutive cropped face frames. An edge map derived from the Laplacian of each frame is used to weight the inter-frame difference, prioritizing structural motion (such as head movement and microexpressions) over flat-region noise. A calibration scalar is applied to boost sensitivity. Static images return 0.0 by design, as no temporal information is available.

Depth Stage (Monocular Depth Estimation)
The depth stage uses the MiDaS small model to estimate a per-frame depth map of the cropped face region. The variance of the predicted depth map is used as a proxy for 3D facial geometry. Real faces exhibit higher depth variance due to natural contours, while flat spoofs produce near-uniform depth. Depth scores are normalized to [0,1].

Fusion and Decision Layer (Rust-Orchestrated)
All four stage outputs are returned to Rust, where a weighted linear fusion is applied. The fused score is mapped to one of three outcomes, Accept, Retry, or Reject, using predefined thresholds. Final system performance is evaluated using False Accept Rate (FAR) and False Reject Rate (FRR), reflecting the asymmetric risk profile of biometric security systems

Results

The multi-stage liveness pipeline was evaluated on 43 samples spanning live and spoof categories (print, cut-out, replay, live image, live video). Fusion weights used: spatial=0.45, texture=0.30, motion=0.25, depth=0.00. Decision thresholds: accept≥0.55, retry≥0.38. Of 36 definitive decisions (excluding retries), the system achieved:

- Definitive accuracy: 83.3% (30/36)
- False Accept Rate (FAR): 0.0%
- False Reject Rate (FRR): 50.0%

Spoof detection performance was strong across all spoof modalities. All printouts, cut-out printouts, and replay samples were either correctly rejected or flagged for retry. No spoof samples were falsely accepted, resulting in a FAR of 0%.

However, performance on genuine live samples was poor. Only 6 live samples were correctly accepted, while 6 were falsely rejected and 6 were flagged for retry. The system struggled particularly with live video inputs, which were almost entirely rejected. Face detection succeeded in 83.9% of frames (1499/1787), with 288 frames falling back to full-frame processing.

Shown below in fig. 1 is the ROC analysis of the fused score across all 43 samples yields an AUC of 0.716, confirming that the multi-stage pipeline produces meaningful separation between live and spoof samples despite using an untrained stub backbone. The curve stays consistently above the random classifier baseline, demonstrating that the combination of spatial, texture, and motion signals carries genuine discriminative signal. The optimal operating point identified by Youden's J statistic occurs at a threshold of 0.436, corresponding to TPR=0.444 and FPR=0.040, meaning that at this threshold the system accepts 44.4% of live users while admitting only 4% of spoofs. The current accept threshold of 0.55 (marked in green) sits to the left of the optimal point, reflecting a conservative security bias that prioritizes FAR=0% at the cost of higher FRR. For a liveness system, false accepts are more damaging than false rejects. The stepped shape of the curve is expected given the small dataset size (43 samples). ROC-based threshold selection will be revisited once the spatial backbone is fine-tuned on anti-spoofing data, at which point the curve is expected to shift toward the upper-left corner and AUC should improve substantially.

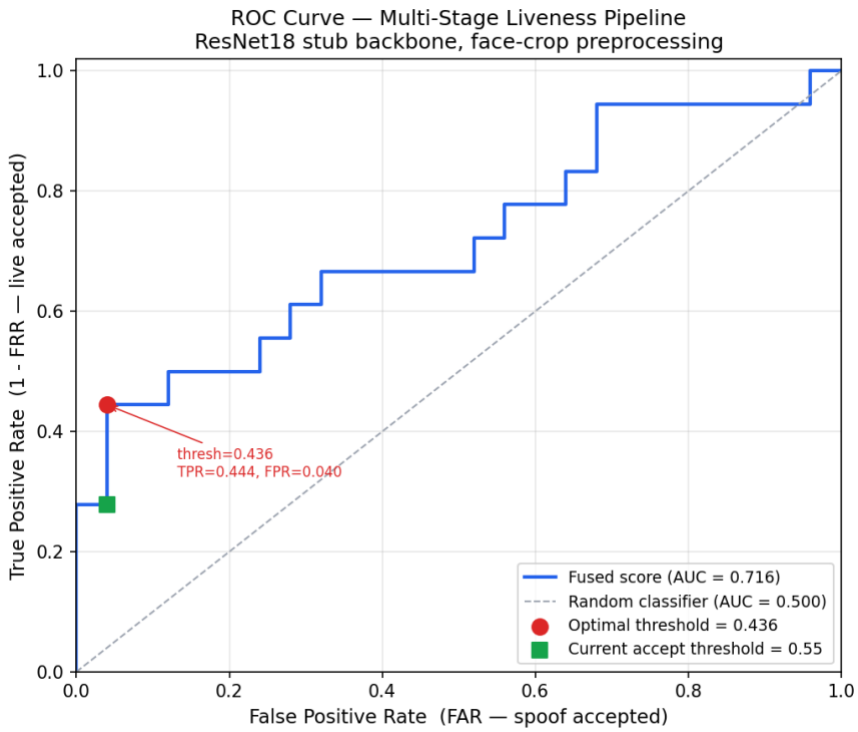

figure 1. *ROC Curve – Multi stage liveness pipeline*

Limitations

The system exhibits strong security bias but poor usability performance. The 0% FAR indicates that the fusion thresholds are highly conservative and effective at preventing spoof acceptance. All spoof categories produced relatively low fused scores, demonstrating that the combination of spatial, texture, and motion signals is sufficient to suppress obvious 2D attacks.

However, the high FRR (50.0%) showed that the model fails to reliably recognize genuine liveness. This is most seen in the live_video category, where motion scores remain low and spatial scores lack discriminative power. Since the spatial backbone (ResNet18 pretrained on ImageNet) is not fine-tuned for anti-spoofing, its outputs reflect image classification confidence rather than liveness-specific features. As a result, live samples often do not receive sufficiently high fused scores to cross the accept threshold.

The motion stage also appears weak. Live video motion values are generally low (0.07–0.19 range), suggesting that simple inter-frame differencing may not sufficiently capture meaningful facial dynamics.

Depth was disabled (weight = 0), removing a potentially discriminative 3D signal from the fusion layer. This likely reduced separation between live and spoof samples in borderline cases. Overall, the system currently behaves as a spoof detector rather than a balanced liveness verifier.

Future work

Future work will focus on improving both discriminative power and operating stability of the system. First, the spatial backbone should be fine-tuned on a dedicated PAD dataset, replacing the current ImageNet-pretrained ResNet18 stub. A liveness-trained model is expected to directly reduce the high False Reject Rate (FRR) by learning spoof-specific features rather than relying on incidental feature separation. Second, depth weighting should be re-enabled and properly calibrated once face cropping is fully validated, allowing the system to exploit true 3D geometric differences between live faces and planar spoof attacks. Third, the current heuristic motion stage should be replaced with learned temporal modeling, such as 3D convolutional networks or temporal pooling architectures, to better capture structured, non-rigid facial dynamics instead of relying on simple frame differencing.

Additionally, the decision thresholds should be recalibrated using ROC analysis to explicitly balance False Accept Rate (FAR) and FRR under defined security requirements, rather than relying on manual tuning. A systematic signal normalization study is also necessary to reassess empirical scaling constants (e.g., Laplacian and depth normalization factors) and ensure that no single stage disproportionately dominates fusion. Finally, detailed error analysis by attack modality should be conducted, particularly for borderline replay cases with high texture or motion scores, to refine feature design and improve robustness across spoof types.